Predicting Body Fat Percentage

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This study investigates whether body fat percentage, a key indicator of health, can be accurately predicted using simple and accessible body measurements. Using a dataset of physical characteristics, a multiple linear regression model was developed to identify the most relevant predictors. Feature selection methods were then applied, and candidate models were evaluated based on out-of-sample performance. The final model, consisting of age, abdomen circumference, wrist circumference, and height, achieved the best balance between predictive accuracy and simplicity. It explains approximately 73% of the variation in body fat percentage. Overall, the findings suggest that body fat percentage can be reliably estimated using a small set of easily obtainable measurements, offering a simple and practical predictive tool.

1. Introduction

Body fat percentage is a key indicator of health that provides more meaningful information than body weight alone. It is commonly used to assess fitness levels and health risks, but accurate measurement typically requires clinical methods such as DEXA scans, which are often costly and not easily accessible.

This study aims to determine whether body fat percentage can be accurately predicted using simple and easily measurable body characteristics. Specifically, it addresses the following questions: how body fat percentage is related to basic body measurement variables; which variables are the most important predictors of body fat percentage; and whether a parsimonious model can achieve strong predictive performance compared to more complex alternatives. These questions are important in assessing whether simple body measurements can serve as a practical and reliable alternative to more complex and costly measurement methods.

To investigate this, we use a dataset of physical characteristics as a training set and evaluate multiple linear regression models to determine which specification provides the best balance of predictive performance and simplicity.

2. Data Set

The bodyfat dataset contains 250 observations of men of various ages. The raw dataset consists of 16 variables, with body fat percentage (pct.bf) as the dependent variable. Aside from density, pct.bf, age, weight, and height, the remaining variables are body circumferences (in centimetres) measuring the perimeter around different body parts. A full summary of the variables is provided in Figure 1 in the appendix.

2.1. Exploratory Data Analysis. From the correlation matrix plot (Figure @ref(fig:corr)), we observe a near-perfect negative correlation (-0.99) between density and body fat percentage. This means we can predict body fat percentage solely from density. However, we note that density is calculated by dividing body mass (weight) by body volume. Although weight is easy to obtain, measuring body volume requires specialised methods such as water displacement, which are costly and not easily accessible. Hence, we decided to drop density from our

Figure 2 also shows a perfect positive correlation between waist and abdomen, indicating that these variables are effectively interchangeable. So we removed waist and retained abdomen.

3. Analysis

3.1. Data Transformation. Height was recorded in inches while all other body measurements (chest, abdomen, and so on) were in centimeters, so height was converted to centimeters for consistency. Weight was converted from pounds to kilograms to align with standard metric units commonly used. We observe that the dependent variable, body fat percentage, has one observation with value 0, which we omitted and identified as measurement error since this is not scientifically possible ().

3.2. Initial Model Experimentation. We start by fitting a linear model on percentage body fat as the dependent variable on all other variables except waist. Plotting residuals against fitted values, the smoother shows a mild wave-like pattern in which residuals are slightly negative at the lower and upper ends of the fitted range, and slightly positive in the middle. However, the deviations are small and the drift at the extremes is likely driven by sparse data rather than genuine nonlinearity. The linearity assumption is therefore considered reasonably satisfied.

The model summary shows a relatively strong fit with an R-squared value of 0.7466 (Adj R-squared: 0.7326), though only age, abdomen, and wrist are statistically significant. Given the strong correlations between predictor variables spotted in EDA, we check VIF for multicollinearity. Weight stands out with a VIF over 40. It is also heavily correlated with other predictors, with over 0.8 correlation with neck, chest, abdomen, hip, thigh, and knee. Beyond this, weight does not distinguish between fat and lean mass, meaning individuals with the same weight can have very different body compositions. As a result, its inclusion may inflate model fit without adding meaningful predictive value, so it is excluded.

Refitting without weight keeps R^2 unchanged at 0.7466 but raises Adj R^2 to 0.7337. Interestingly, neck and height both become statistically significant - previously, weight was absorbing their contribution due to high correlation, potentially obscuring their association with body fat. Checking VIF again, chest, abdomen, hip, and thigh still show VIF > 5. This is expected because chest and abdomen both measure trunk (torso) size, and hip and thigh both capture lower body size, so the variables are essentially telling the same story. We retain the stronger representative from each pair to address multicollinearity. Abdomen is kept over chest as it is more strongly correlated to body fat percentage, and hip is kept over thigh for similar reasons. Chest and thigh are dropped on both grounds of high VIF and statistical insignificance ($p > 0.10$).

However, upon refitting, hip continues to show high VIF with abdomen and remains statistically insignificant. With no statistical or unique theoretical justification to retain it, hip is dropped. Knee and ankle are removed at this stage too. Intuitively, these are joints so their size is largely determined by bone structure, and unlike the abdomen or hip, these sites are less directly associated

with fat accumulation. This intuition is reflected in their very high p-values, confirming they add little predictive value. Bicep and forearm are also statistically insignificant. This is not surprising because arm circumferences are heavily influenced by muscle mass not fat. A person with well-developed arms from regular physical activity would have large bicep and forearm measurements despite potentially low body fat, meaning these variables conflate fat and lean mass in a way that weakens their predictive value. Neck, while borderline significant here ($p = 0.058$), becomes insignificant once bicep and forearm are removed, suggesting its earlier significance was partly a result of the surrounding noise. All three are dropped, giving our first candidate model (age + abdomen + wrist + height_cm).

3.3. Model Comparison. After manually arriving at a candidate model, we also identified candidate models through three feature selection approaches:

3.3.1. Exhaustive Search. We first perform an exhaustive search, which considers all possible combinations of predictors. Each model is evaluated using BIC (Bayesian Information Criterion), which, compared to the Akaike Information Criterion, imposes a heavier penalty on model complexity and therefore tends to favour simpler, more parsimonious models. As this aligns with our objective, we adopt BIC as our selection criterion.

We observe that BIC reaches its minimum when the model includes 4 predictors. Therefore, based on Figure X, we select the model with age, height, abdomen, and wrist. Interestingly, we obtain the same model as the first candidate model.

3.3.2. Backward Variable Selection. We also conducted backward variable selection based on BIC. In this approach, variables are removed one at a time until the model reaches its minimum BIC value. Based on Figure X, the model achieves the lowest BIC when age, height, abdomen, and wrist are included as predictors. Notably, this also results in the same set of predictors as those obtained from the manual and exhaustive search methods.

3.3.3. Forward Variable Selection. For further comparison, we also performed forward variable selection based on BIC. In this approach, variables are added one at a time until the model reaches its minimum BIC value. The resulting model includes abdomen, weight, and wrist as predictors, which becomes our second candidate model.

To assess which candidate model is best, we perform out-of-sample tests next.

3.4. Cross Validation. To evaluate the out-of-sample performance on unseen data, we apply 10-fold cross-validation across our 2 candidate models. Rather than a single train-test split, 10-fold cross-validation repeatedly partitions the data into 10 equal folds, training on 9 and testing on the remaining 1, cycling through all folds. This gives a more robust estimate of out-of-sample performance for a dataset of this size.

We tested this approach on 2 different models, with the differences as listed below:

- Model 1: Manual/Exhaustive search model/Backward variable selection
- Model 2: Forward variable selection

Both models perform very similarly in terms of predictive ability.

Based on MAE and RMSE (see Figure X), Model 1 performs slightly better than Model 2, with MAE lower by 0.034 and RMSE lower by 0.031. This suggests that Model 1 produces more accurate predictions on average and is slightly better at limiting large prediction errors.

In contrast, Model 2 has a marginally higher R2 (0.745 vs 0.724), indicating that it explains a slightly greater proportion of the variation in body fat percentage. However, a higher R2 does not necessarily imply better predictive performance, especially in this case. The improvement in R2 is likely driven by the inclusion of weight, which we identified as a problematic variable due to multicollinearity and weak conceptual relevance.

Importantly, both exhaustive and backward selection procedures (based on BIC) converge to the same specification as Model 1. This provides additional evidence that a parsimonious specification can achieve comparable, if not better, predictive performance. Therefore, Model 1 is preferred as the more parsimonious and robust model, retaining only the most reliable predictors while avoiding unnecessary complexity.

3.5. Assumptions Checking. In Figure X, the residuals vs fitted values plot for the final model shows a mild wave-like pattern, similar to Figure X for the initial model. However, as discussed previously, this is likely driven by limited observations at the upper extremes rather than true non-linearity, so the linearity assumption is considered satisfied. Residual diagnostics further show that the Q-Q plot is approximately linear, suggesting no major violations of normality. The scale-location plot shows a roughly horizontal trend, indicating no clear evidence of heteroskedasticity. Finally, independence is assumed since each observation corresponds to a different individual, with no repeated measurements, clustering, or time-series structure in the data.

Finally, no observations exceed three standard deviations from the mean, suggesting that there are no significant outliers.

4. Discussion and Conclusion

$$pct.bf = 3.69 + 0.056age + 0.767abdomen - 1.929wrist - 0.128height_cm \quad [1]$$

5. References

A. Appendix

Figures and Tables should be labelled and referenced in the standard way using the `\label{}` and `\ref{}` commands.

```
\begin{table}

\caption{\label{tab:figure1}Summary of variables and units}
\centering
\begin{tabular}[t]{lll}
\toprule
Variable & Description & Unit\\
\midrule
Density & Body density & g/cm³\\
Pct.BF & Body fat percentage & \%\\
Age & Age of individual & years\\
Weight & Body weight & lb\\
Height & Height & in\\
\addlinespace
Neck & Neck circumference & cm\\
Chest & Chest circumference & cm\\
Abdomen & Abdomen circumference & cm\\
Waist & Waist circumference & in\\
Hip & Hip circumference & cm\\
\addlinespace
Thigh & Thigh circumference & cm\\
Knee & Knee circumference & cm\\
Ankle & Ankle circumference & cm\\
Bicep & Bicep circumference & cm\\
Forearm & Forearm circumference & cm\\
\addlinespace
Wrist & Wrist circumference & cm\\
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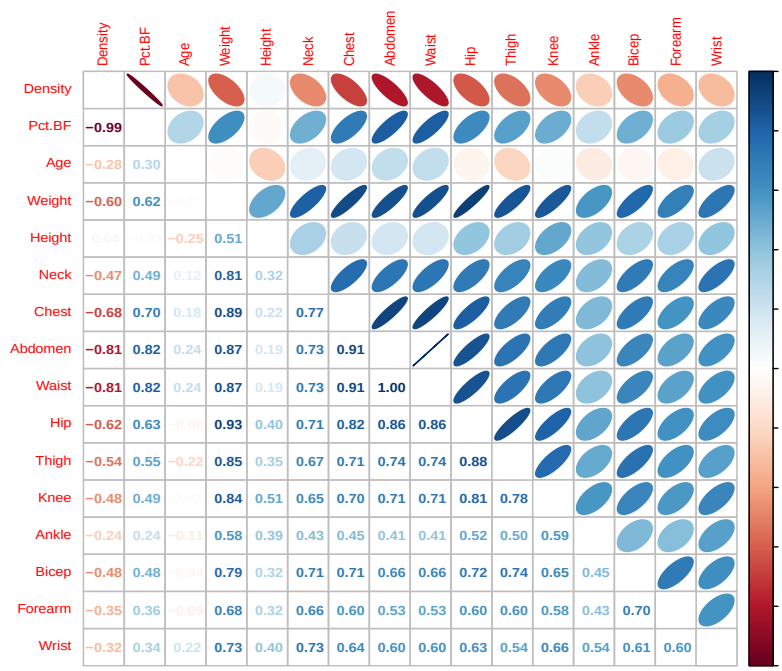

Fig. 1. Correlation Matrix Plot

References